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Patient support apparatus and headwall unit syncing

Abstract

A wireless communication system for exchanging messages between a patient support apparatus and a headwall unit includes a syncing function wherein certain settings that are present in the headwall unit before a communication link is established with the patient support apparatus are synced with values for those settings that come from the patient support apparatus after the communication link is initially established. Similarly, certain settings that are present in the patient support apparatus before the communication link is established are synced with values for those settings that come from the headwall unit after the communication link is initially established. The syncing function thereby ensures that, upon the initial establishment of the communication link, the patient support apparatus settings will match the headwall settings, and vice versa.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The application claims priority to and all advantages of U.S. Provisional Patent Application No. 63/193,778 filed on 27 May 2022, the content of which is incorporated herein by reference.

BACKGROUND

(1) The present disclosure relates to patient support apparatuses, such as beds, cots, stretchers, recliners, or the like, that are adapted to communicate with an existing nurse call system and/or one or more room devices.

(2) Existing hospital beds often communicate with a nurse call system over a communication link. In some situations where the communication link is a wireless communication link, the hospital bed may communicate with a headwall unit attached to a headwall of a healthcare facility room. The headwall unit acts as an intermediary device between the bed and a wall outlet of the nurse call system. The headwall unit therefore forwards communications received from the bed to the nurse call system wall outlet, and forwards communications received from the nurse call system wall outlet to the bed.

SUMMARY

(3) According to various embodiments, the present disclosure provides an improved system for helping to ensure that one or more conditions onboard a patient support apparatus, such as a bed, are synced with one or more corresponding conditions onboard the headwall unit, and vice versa, thereby helping to ensure that errors and/or other undesirable conditions do not take place in response to a mismatch between the bed conditions and the headwall conditions. The patient support apparatuses and headwall units disclosed herein are therefore configured to effectuate a syncing process that ensures that some of the bed conditions are synced with some of the headwall conditions, and some of the headwall conditions are synced with the bed conditions when the communication link is initially established.

(4) According to a first embodiment of the present disclosure, a headwall unit is provided that is adapted to be mounted to a headwall of a room in a healthcare facility and used as part of the communication link between a patient support apparatus and a nurse call system. The headwall unit includes a nurse call interface, a first wireless transceiver, and a controller. The nurse call interface is adapted to electrically couple to a wall outlet mounted in the headwall, and it includes a set of pins adapted to electrically couple to a plurality of conductors defined in the wall outlet when the nurse call interface is coupled to the wall outlet. Each of the pins are adapted to be in different electrical states. The first wireless transceiver is adapted to wirelessly communicate with a second wireless transceiver positioned onboard the patient support apparatus when the patient support apparatus is positioned adjacent to the headwall unit. The controller is adapted to use the first wireless transceiver to establish a communication link with the patient support apparatus. In response to the communication link being established, the controller is configured to perform the following: (a) receive a plurality of state values from the patient support apparatus; and sync the electrical states of a subset of the set of pins to match the state values received from the patient support apparatus.

(5) According to another aspect of the present disclosure, a first one of the state values received from the patient support apparatus defines an on or off state of a reading light. The reading light is adapted to provide illumination to a patient on the patient support apparatus.

(6) In some embodiments, a second one of the state values received from the patient support apparatus defines an on or off state of a room light. The room light is adapted to provide

illumination to an entire room in which the headwall unit it positioned.

(7) In some embodiments, a first one of the state values received from the patient support apparatus defines an on or off state of a nurse call request. The nurse call request is adapted to notify a nurse call system coupled to the wall outlet of a patient's desire to speak with a nurse.

(8) In some embodiments, a first one of the state values received from the patient support apparatus defines an on or off state of a priority alert signal. The priority alert signal is adapted to notify a nurse call system of a priority event occurring on the patient support apparatus.

(9) In some embodiments, a first one of the state values received from the patient support apparatus defines an on or off state of a television control signal, and the television control signal is adapted to instruct a television to be turned on or off.

(10) The headwall unit, in some embodiments, further comprises a third wireless transceiver adapted to wirelessly communicate with a fourth wireless transceiver positioned onboard the patient support apparatus when the patient support apparatus is positioned adjacent to the headwall unit.

(11) In some embodiments, the communication link is established when both the first wireless transceiver and the third wireless transceiver are in communication with the second and fourth wireless transceivers, respectively, onboard the patient support apparatus.

(12) In some embodiments, the first wireless transceiver is a radio frequency transceiver and the third wireless transceiver is an infrared transceiver.

(13) The controller, in some embodiments, is further adapted to send a second plurality of state values to the patient support apparatus in response to the communication link being established. The second plurality of state values correspond to a current electrical state of a second subset of the set of pins that is different from the subset.

(14) The controller, in some embodiments, is further adapted to receive audio signals from the patient support apparatus via the first wireless transceiver and to forward the audio signals to at least one of the pins.

(15) The controller, in some embodiments, is further adapted to receive an exit alert message from the patient support apparatus and to change an electrical state of at least one of the pins in response to receiving the exit alert message.

(16) In some embodiments, the headwall unit further includes a memory in which is stored a unique identifier that uniquely identifies the headwall unit, and the controller is further adapted to forward the unique identifier to the patient support apparatus using the first wireless transceiver.

(17) The controller, in some embodiments, is further adapted to receive a configuration message via the first wireless transceiver. The configuration message includes an identification of a first pair of the pins that must be electrically shorted together to avoid triggering a cord-out alert in a nurse call system coupled to the wall outlet.

(18) The controller, in some embodiments, is further adapted to receive a new configuration message via the first wireless transceiver which includes a new identification of a second pair of pins that must be electrically shorted together to avoid triggering the cord-out alert. The second pair of pins is different from the first pair of pins.

(19) According to another embodiment of the present disclosure, a patient support apparatus is provided. The patient support apparatus includes a support surface, a first wireless transceiver, and a controller. The support surface is adapted to support a patient thereon. The first wireless transceiver is adapted to wirelessly communicate with a second wireless transceiver positioned onboard a headwall unit mounted to a headwall of room in a healthcare facility. The controller is adapted to use the first wireless transceiver to establish a communication link with the headwall unit and to perform the following in response to the communication link being established: (a) receive a plurality of headwall state values from the headwall unit; and (b) sync a plurality of patient support apparatus state values to match the headwall state values received from the headwall unit.

- (20) According to another aspect of the present disclosure, the patient support apparatus further comprises a nurse answer light adapted to be illuminated when a remotely positioned nurse answers a call from the patient, and a first one of the headwall state values defines an on or off state of the nurse answer light.
- (21) In some embodiments, a nurse call light is adapted to be illuminated when the patient calls a remotely positioned nurse, and a first one of the headwall state values defines an on or off state of the nurse call light.
- (22) In some embodiments, the patient support apparatus further comprises a cable detector adapted to detect a presence of a cable connecting the patient support apparatus to a wall outlet defined in the headwall, and a first one of the headwall state values defines an enabled or disabled state of the cable detector.
- (23) In some embodiments, the patient support apparatus further comprises a memory in which is stored a configuration setting of the headwall unit, and a first one of the headwall state values defines a configured state or an unconfigured state of the headwall unit.
- (24) In some embodiments, the patient support apparatus further comprises a memory in which is stored a television setting of a television coupled to the headwall unit, and a first one of the headwall state values defines a type of the television.
- (25) In some embodiments, the patient support apparatus further comprises a memory in which is stored a cable setting of a cable extending between the headwall unit and a wall outlet defined in the headwall, and a first one of the headwall state values defines a connected state or an unconnected state of the cable.
- (26) In some embodiments, the patient support apparatus further comprises a memory in which is stored an interlock setting of a pair of pins of the headwall unit, and a first one of the headwall state values defines an electrically shorted state or unshorted state of the pair of pins.
- (27) In some embodiments, the patient support apparatus further comprises a third wireless transceiver adapted to wirelessly communicate with a fourth wireless transceiver positioned onboard the headwall unit when the patient support apparatus is positioned adjacent to the headwall unit.
- (28) The communication link, in some embodiments, is established when both the first wireless transceiver and the third wireless transceiver are in communication with the second and fourth wireless transceivers, respectively, onboard the headwall unit.
- (29) In some embodiments, the first wireless transceiver is a radio frequency transceiver and the third wireless transceiver is an infrared transceiver.
- (30) The controller, in some embodiments, is further adapted to send audio signals to the headwall unit via the first wireless transceiver that are to be forwarded by the headwall unit to a wall outlet integrated into the headwall.
- (31) In some embodiments, the patient support apparatus further comprises an exit detection system adapted to detect a patient exit from the support surface. In such embodiments, the controller may be further adapted to send an exit alert message to the headwall unit via the first wireless transceiver that is to be forwarded by the headwall unit to a wall outlet integrated into the headwall.
- (32) The patient support apparatus, in some embodiments, further comprises a memory in which is stored a first unique identifier that uniquely identifies the patient support apparatus; and a network transceiver adapted to communicate with a computer network of the healthcare facility. In such embodiments, the controller may be further adapted to receive a second unique identifier of the headwall unit from the headwall unit and to forward both the first and second unique identifiers to a server on the computer network via the network transceiver.
- (33) In some embodiments, the patient support apparatus is a bed or a stretcher that includes a plurality of wheels, a plurality of siderails, and a height-adjustable litter frame. Still further, in some embodiments, the sync described herein takes places immediately after the communication link is established, such as within a second or two, although other time frames may be used.

(34) Before the various embodiments disclosed herein are explained in detail, it is to be understood that the claims are not to be limited to the details of operation or to the details of construction and the arrangement of the components set forth in the following description or illustrated in the drawings. The embodiments described herein are capable of being practiced or being carried out in alternative ways not expressly disclosed herein. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of “including” and “comprising” and variations thereof is meant to encompass the items listed thereafter and equivalents thereof as well as additional items and equivalents thereof. Further, enumeration may be used in the description of various embodiments. Unless otherwise expressly stated, the use of enumeration should not be construed as limiting the claims to any specific order or number of components. Nor should the use of enumeration be construed as excluding from the scope of the claims any additional steps or components that might be combined with or into the enumerated steps or components.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a patient support apparatus according to a first embodiment of the present disclosure;
- (2) FIG. 2 is a perspective view of a litter frame and a pair of lift header assemblies of the patient support apparatus;
- (3) FIG. 3 is a perspective view of a base and a pair of lifts of the patient support apparatus;
- (4) FIG. 4 is a diagram of the patient support apparatus, a wireless headwall unit, and a wireless communication link between the patient support apparatus and a wall outlet of a nurse call system;
- (5) FIG. 5 is a diagram of an alternative wireless headwall unit;
- (6) FIG. 6 is a block diagram of several of the structures of FIG. 4 showing some of the internal components of the headwall unit and the patient support apparatus;
- (7) FIG. 7 is a plan view of an illustrative patient control panel of the patient support apparatus;
- (8) FIG. 8 is a table of settings for the patient support apparatus and headwall unit that are synced shortly after the communication link is established between the patient support apparatus and headwall unit;
- (9) FIG. 9 is a perspective view of a prior art 37-pin male cable connector;
- (10) FIG. 10 is a perspective view of a prior art 37-pin female cable connector; and
- (11) FIG. 11 is a chart of a prior art example of the functions of the pins of a 37-pin cable often used in existing healthcare facilities.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(12) An illustrative patient support apparatus **20** according to a first embodiment of the present disclosure is shown in FIG. 1. Although the particular form of patient support apparatus **20** illustrated in FIG. 1 is a bed adapted for use in a hospital or other medical setting, it will be understood that patient support apparatus **20** could, in different embodiments, be a cot, a stretcher, a recliner, a chair, or any other patient support structure that communicates with a nurse call outlet of a conventional nurse call system.

(13) In general, patient support apparatus **20** includes a base **22** having a plurality of wheels **24**, a pair of lifts **26** supported on the base **22**, a litter frame **28** supported on the lifts **26**, and a support deck **30** supported on the litter frame **28**. Patient support apparatus **20** further includes a footboard **32** and a plurality of side rails **34**. Side rails **34** are all shown in a raised position in FIG. 1 but are each individually movable to a lower position in which ingress into, and egress out of, patient support apparatus **20** is not obstructed by the lowered side rails **34**.

(14) Lifts **26** are adapted to raise and lower litter frame **28** with respect to base **22**. Lifts **26** may be

hydraulic actuators, pneumatic actuators, electric actuators, or any other suitable device for raising and lowering litter frame **28** with respect to base **22**. In the illustrated embodiment, lifts **26** are operable independently so that the tilting of litter frame **28** with respect to base **22** can also be adjusted. That is, litter frame **28** includes a head end **36** and a foot end **38**, each of whose height can be independently adjusted by the nearest lift **26**. Patient support apparatus **20** is designed so that when an occupant lies thereon, his or her head will be positioned adjacent head end **36** and his or her feet will be positioned adjacent foot end **38**.

(15) Litter frame **28** provides a structure for supporting support deck **30**, footboard **32**, and side rails **34**. Support deck **30** provides a support surface for a mattress (not shown in FIG. **1**), such as, but not limited to, an air, fluid, or gel mattress. Alternatively, another type of soft cushion may be supported on support deck **30** so that a person may comfortably lie and/or sit thereon. The top surface of the mattress or other cushion forms a support surface for the occupant.

(16) Support deck **30** is made of a plurality of sections, some of which are pivotable about generally horizontal pivot axes. In the embodiment shown in FIG. **1**, support deck **30** includes a head section **40**, a seat section **42**, a thigh section **44**, and a foot section **46**. Head section **40**, which is also sometimes referred to as a Fowler section, is pivotable about a generally horizontal pivot axis between a generally horizontal orientation (not shown in FIG. **1**) and a plurality of raised positions (one of which is shown in FIG. **1**). Thigh section **44** and foot section **46** may also be pivotable about generally horizontal pivot axes.

(17) Patient support apparatus **20** further includes a plurality of control panels **48** that enable a user of patient support apparatus **20**, such as a patient and/or an associated caregiver, to control one or more aspects of patient support apparatus **20**. In the embodiment shown in FIG. **1**, patient support apparatus **20** includes a footboard control panel **48a**, a pair of outer side rail control panels **48b** (only one of which is visible), and a pair of inner side rail control panels **48c** (only one of which is visible). Footboard control panel **48a** and outer side rail control panels **48b** are intended to be used by caregivers, or other authorized personnel, while inner side rail control panels **48c** are intended to be used by the patient associated with patient support apparatus **20**. Each of the control panels **48** includes a plurality of controls **126** (FIG. **6**), although each control panel **48** does not necessarily include the same controls and/or functionality. FIG. **7** illustrates one example of a patient control panel **48c**, although it will be understood that other types patient control panels **48c** may be used.

(18) Among other functions, controls **126** (FIG. **6**) of control panel **48a** allow a user to control one or more of the following: change a height of litter frame **28**, raise or lower head section **40**, activate and deactivate a brake for wheels **24**, arm and disarm an exit detection system, etc. One or both of the inner siderail control panels **48c** also include at least one control that enables a patient to call a remotely located nurse (or other caregiver). In addition to the nurse call control, one or both of the inner siderail control panels **48c** also include one or more controls for controlling one or more features of one or more room devices positioned within the same room as the patient support apparatus **20**. As will be described in more detail below, such room devices include, but are not necessarily limited to, a television, a reading light, and/or a room light. With respect to the television, the features that may be controllable by one or more controls on control panel **48c** include, but are not limited to, the volume, the channel, the closed-captioning, and/or the power state of the television. With respect to the room and/or night lights, the features that may be controlled by one or more controls on control panel **48c** include the on/off state and/or the brightness level of these lights.

(19) Footboard control panel **48a** is implemented in the embodiment shown in FIG. **1** as a control panel having a lid (flipped down in FIG. **1**) underneath which is positioned a plurality of controls. As with all of the controls of the various control panels **48**, the controls of control panel **48a** may be implemented as buttons, dials, switches, or other devices. Any of control panels **48a-c** may also include a display for displaying information regarding patient support apparatus **20**. The display is a touchscreen in some embodiments.

(20) In some embodiments, footboard control panel **48a** may take on the form of the footboard control panel **54a** disclosed in commonly assigned PCT patent application serial number PCT/US2021/32426 filed May 14, 2021, by applicant Stryker Corporation and entitled PATIENT SUPPORT APPARATUSES WITH HEADWALL COMMUNICATION, the complete disclosure of which is incorporated herein by reference. Additionally, or alternatively, patient control panel **48c** may take on the form of the patient control panel **54c** disclosed in the aforementioned PCT patent application. Other types of footboard control panels **48a-c** may, of course, be implemented.

(21) FIG. 2 illustrates in greater detail litter frame **28** separated from lifts **26** and base **22**. Litter frame **28** is also shown in FIG. 2 with support deck **30** removed. Litter frame **28** is supported by two lift header assemblies **50**. A first one of the lift header assemblies **50** is coupled to a top **52** (FIG. 3) of a first one of the lifts **26**, and a second one of the lift header assemblies **50** is coupled to the top **52** of the second one of the lifts **26**. Each lift header assembly **50** includes a pair of force sensors **54**, which will be described herein as being load cells, but it will be understood that force sensors **54** may be other types of force sensors besides load cells. The illustrated embodiment of patient support apparatus **20** includes a total of four load cells **54**, although it will be understood by those skilled in the art that different numbers of load cells may be used in accordance with the principles of the present disclosure. Load cells **54** are configured to support litter frame **28**. More specifically, load cells **54** are configured such that they provide complete and exclusive mechanical support for litter frame **28** and all of the components that are supported on litter frame **28** (e.g. support deck **30**, footboard **32**, side rails **34**, etc.). Because of this construction, load cells **54** are adapted to detect the weight of not only those components of patient support apparatus **20** that are supported by litter frame **28** (including litter frame **28** itself), but also any objects or persons who are wholly or partially being supported by support deck **30**.

(22) The mechanical construction of patient support apparatus **20**, as shown in FIGS. 1-3, is the same as, or nearly the same as, the mechanical construction of the Model 3002 S3 bed manufactured and sold by Stryker Corporation of Kalamazoo, Michigan. This mechanical construction is described in greater detail in the Stryker Maintenance Manual for the MedSurg Bed, Model 3002 S3, published in 2010 by Stryker Corporation of Kalamazoo, Michigan, the complete disclosure of which is incorporated herein by reference. It will be understood by those skilled in the art that patient support apparatus **20** can be designed with other types of mechanical constructions, such as, but not limited to, those described in commonly assigned, U.S. Pat. No. 7,690,059 issued to Lemire et al., and entitled HOSPITAL BED; commonly assigned U.S. Pat. Publication No. 2007/0163045 filed by Becker et al. and entitled PATIENT HANDLING DEVICE INCLUDING LOCAL STATUS INDICATION, ONE-TOUCH FOWLER ANGLE ADJUSTMENT, AND POWER-ON ALARM CONFIGURATION; and/or commonly assigned, U.S. Pat. No. 10,130,536 to Roussy et al., entitled PATIENT SUPPORT USABLE WITH BARIATRIC PATIENTS, the complete disclosures of all of which are also hereby incorporated herein by reference. The mechanical construction of patient support apparatus **20** may also take on forms different from what is disclosed in the aforementioned references.

(23) Load cells **54** are part of an exit detection system **56** (FIG. 6) that, when armed, issues an alert when the patient exits from patient support apparatus **20**. Exit detection system **56** is adapted to be armed via control panel **48a**. After being armed, exit detection system **56** determines when an occupant of patient support apparatus **20** has left, or is likely to leave, patient support apparatus **20**, and issues an alert and/or notification to appropriate personnel so that proper steps can be taken in response to the occupant's departure (or imminent departure) in a timely fashion. In at least one embodiment, exit detection system **56** monitors the center of gravity of the patient using the system and method disclosed in commonly assigned U.S. Pat. No. 5,276,432 issued to Travis and entitled PATIENT EXIT DETECTION MECHANISM FOR HOSPITAL BED, the complete disclosure of which is incorporated herein by reference. In other embodiments, exit detection system **56** determines if the occupant is about to exit, or already has exited, from patient support apparatus **20**.

by determining a distribution of the weights detected by each load cell **54** and comparing the detected weight distribution to one or more thresholds. In such embodiments, the center of gravity may or may not be explicitly calculated.

(24) Other manners for functioning as an exit detection system are also possible. These include, but are not limited to, any of the manners disclosed in the following commonly assigned patent applications: U.S. patent application Ser. No. 14/873,734 filed Oct. 2, 2015, by inventors Marko Kostic et al. and entitled PERSON SUPPORT APPARATUS WITH MOTION MONITORING; U.S. patent publication 2016/0022218 filed Mar. 13, 2014, by inventors Michael Hayes et al. and entitled PATIENT SUPPORT APPARATUS WITH PATIENT INFORMATION SENSORS; and U.S. patent application Ser. No. 15/266,575 filed Sep. 15, 2016, by inventors Anuj Sidhu et al. and entitled PERSON SUPPORT APPARATUSES WITH EXIT DETECTION SYSTEMS, the complete disclosures of all of which are incorporated herein by reference. Further, in some embodiments, load cells **54** may be part of both an exit detection system and a scale system that measures the weight of a patient supported on support deck **30**. The outputs from the load cells **54** are processed, in some embodiments, in any of the manners disclosed in commonly assigned U.S. patent application Ser. No. 62/428,834 filed Dec. 1, 2016, by inventors Marko Kostic et al. and entitled PERSON SUPPORT APPARATUSES WITH LOAD CELLS, the complete disclosure of which is incorporated herein by reference.

(25) Regardless of how implemented, patient support apparatus **20** is adapted to communicate an alert when the exit detection system is armed and detects that a patient is about to, or has, exited. One manner in which the alert is communicated to a conventional nurse call system **68** is shown in FIG. **4**. Patient support apparatus **20** communicates with a nurse call system **68**, and vice versa, through a communication link **58** that is established between patient support apparatus **20** and a conventional nurse call wall outlet **60**. Communication link **58** is a wireless communication link. In addition to forwarding an exit detection alert over communication link **58** from patient support apparatus **20** to nurse call system **68**, communication link **58** may be used for communicating a variety of other information.

(26) One example of such information are the audio signals of the patient and a remotely positioned nurse. That is, a patient onboard patient support apparatus **20** is able to communicate with a remotely positioned nurse by speaking into a microphone onboard patient support apparatus **20**, and patient support apparatus **20** forwards these audio signals to a remotely positioned nurse by transmitting them over communication link **58** to wall outlet **60**, which is in communication with nurse call system **68**, as will be discussed in greater detail below. Similarly, a remotely positioned nurse is able to speak into a microphone coupled to the nurse call system and have his/her voice signals forwarded to wall outlet **60**, which are then transmitted over communication link **58** to a speaker onboard patient support apparatus **20**.

(27) FIG. **4** illustrates additional details of a typical healthcare facility **64**. As shown therein, healthcare facility **64** includes a headwall **66**, nurse call system **68**, a plurality of rooms **70** (**70a**, **70b** . . . **70x**), one or more nurses' stations **72**, a local area network **74**, one or more wireless access points **76**, a patient support apparatus server **78**, and one or more network appliances **80** that couple LAN **74** to the internet **82**, thereby enabling server **78** and other applications on LAN **74** to communicate with computers outside of healthcare facility **64**, such as, but not limited to, a geographically remote server **84**. Wall outlet **60** is typically electrically coupled by one or more conductors **92** to a television **86** and one or more room devices (e.g. a room light **62a**, a reading light **62b**, etc.). It will be understood by those skilled in the art, however, that the healthcare facility infrastructure shown in FIG. **4** may vary widely from healthcare facility to healthcare facility.

(28) For example, patient support apparatus **20** may be used in healthcare facilities having no wireless access points **76**, no connection to the internet **82** (e.g. no network appliances **80**), and/or no patient support apparatus server **78**. Still further, local area network **74** may include other and/or additional servers installed thereon, and/or room **70**, in some healthcare facilities **64**, may be semi-

private room having multiple patient support apparatuses **20** and multiple wall outlets **60**. Still other variations are possible. It will therefore be understood that the particular healthcare facility infrastructure shown in FIG. **4** is merely illustrative, and that patient support apparatus **20** is constructed to be communicatively coupled to healthcare facility communication infrastructures which are arranged differently from that of FIG. **4**, some of which are discussed in greater detail below.

(29) As is shown in FIG. **4**, patient support apparatus **20** is adapted to be communicatively coupled to the wall outlet **60** on headwall **66** by way of a wireless communication link **58** that wirelessly couples patient support apparatus **20** to a wireless headwall unit **94**. Headwall unit **94**, in turn, is coupled by way of a cable **90** to wall outlet **60**. Headwall unit **94** and patient support apparatus **20** are able to communicate wirelessly with each other in a bidirectional fashion. That is, messages can be wirelessly sent from patient support apparatus **20** to headwall unit **94**, and messages can be wirelessly sent from headwall unit **94** to patient support apparatus **20**.

(30) Although not shown in FIG. **4**, patient support apparatus **20** may further be configured to be able to communicate with wall outlet **60** via a cable, if desired. When such wired communication is desired, a nurse call cable **90** is connected directly from patient support apparatus **20** to wall outlet **60**. Unless wall outlet **60** has room for two cable plugs, the end of cable **90** shown in FIG. **4** that is plugged into wall outlet **60** is removed in order to allow the cable from patient support apparatus **20** to be inserted therein. Alternatively, the end of the cable **90** shown in FIG. **4** that is plugged into headwall unit **94** may be removed and plugged into patient support apparatus **20**. However attained, once a cable **90** is coupled between patient support apparatus **20** and wall outlet **60**, a wired connection is established that bypasses headwall unit **94** (if present).

(31) Wall outlet **60** is coupled to one or conductors **92** that electrically couple the wall outlet **60** to nurse call system **68** and to one or more other devices, such as television **86**, room light **62a**, and/or reading light **62b**. Conductors **92** are typically located behind headwall **66** and not visible. In some healthcare facilities, conductors **92** may first couple to a room interface board that includes one or more electrical connections electrically coupling the room interface board to television **86** and/or nurse call system **68**. Still other communicative arrangements for coupling wall outlet **60** to nurse call system **68** and television **86** are possible.

(32) Communication link **58** (FIG. **4**) enables patient support apparatus **20** to communicate with nurse call system **68**, television **86**, room light **62a**, and/or reading light **62b**. A patient supported on patient support apparatus **20** who activates a nurse call control on patient support apparatus **20** causes a signal to be conveyed via communication link **58** to the nurse call system **68**, which then sends a notification to one or more remotely located nurses (e.g. nurses at one of the nurses' stations **72**). If the patient uses a TV control positioned on one of the control panels (e.g. control **126k** of control panel **48c**; see FIG. **7**) to change a channel or change the volume of television **86**, the control conveys a signal along link **58** to the wall outlet **60**, and the signal is thereafter passed from outlet **60** to television **86**. Similarly, if the patient uses a room light or reading light control on one of the control panels, he or she is able to turn on or off the room light **62a** and reading light **62b**.

(33) As will be discussed in greater detail below, outlet **60** often includes a plurality of pins (e.g. 37 pins), and the audio signals that are passed between the patient and a remotely positioned nurse are transmitted over a separate set of pins than the pins that transmits control signals for controlling television **86**. Additional pins are used for communicating other information between patient support apparatus **20** and nurse call system **68** and/or other devices positioned within room **70** (e.g. television **86**, room light **62a**, reading light **62b**).

(34) FIG. **5** illustrates an alternative embodiment of a headwall unit **94a** according to another embodiment of the present disclosure. Headwall unit **94a** differs from headwall unit **94** in that, rather than including a connector for a cable **90**, it is adapted to plug directly into wall outlet **60** via a connector **102**. Connector **102** is thus shaped and dimensioned to be frictionally maintained in an

electrically coupled state to outlet **60**, and to support the entire headwall unit **94a**. One or more alignment posts **104** may be included with connector **102** in order to more securely retain headwall unit **94a** to wall outlet **60**, if desired. Connector **102** may be the same as, or nearly the same as, an end of a cable **90**, thereby allowing either a cable from headwall unit **94**, or connector **102** from headwall unit **94a**, to be inserted into wall outlet **60**.

(35) In the embodiment shown in FIG. 5, connector **102** is a 37 pin connector that includes 37 pins adapted to be inserted into 37 mating sockets of wall outlet **60**. Such 37 pin connections are one of the most common types of connectors found on existing headwalls of medical facilities for making connections to the nurse call system **68**. Such 37 pin connectors, however, are not the only type of connectors, and it will be understood that headwall unit **94a** can utilize different types of connectors **102** (whether integrated therein or attached to a cable) that are adapted to electrically couple to different types of nurse call outlets **60**. One example of such an alternative wall outlet **60** and cable is disclosed in commonly assigned U.S. patent application Ser. No. 14/819,844 filed Aug. 6, 2015, by inventors Krishna Bhimavarapu et al. and entitled PATIENT SUPPORT APPARATUSES WITH WIRELESS HEADWALL COMMUNICATION, the complete disclosure of which is incorporated herein by reference. Still other types of wall outlets **60** and corresponding cable connectors **102** may be utilized.

(36) Other than the absence of a connector for a cable **90**, headwall unit **94a** may be the same as headwall unit **94**, and the following description of headwall unit **94** will apply equally to both headwall unit **94** and headwall unit **94a**. Headwall unit **94** of FIG. 4 (and headwall unit **94a** of FIG. 5) include a power cable **106** having an end adapted to be inserted into a conventional electrical outlet **108**. Power cable **106** enables headwall unit **94** to receive power from the mains electrical supply via outlet **108**. It will be appreciated that, in some embodiments, headwall unit **94** is battery operated and cable **106** may be omitted. In still other embodiments, headwall unit **94** may be both battery operated and include cable **106** so that in the event of a power failure, battery power supplies power to headwall unit **94**, and/or in the event of a battery failure, electrical power is received through outlet **108**.

(37) Headwall unit **94** may also include a plurality of status lights **110**, such as are shown in FIG. 5. Status lights **110** provide visual indications about one or more aspects of headwall unit **94**. For example, in some embodiments, the illumination of one of status lights **110** indicates that headwall unit **94** is in successful communication with nurse call system **68** and/or patient support apparatus **20**. The illumination of one or more additional status lights **110** may also or alternatively indicate that power is being supplied to headwall unit **94** and/or the status of a battery included within headwall unit **94**. Still further, in some embodiments, one or more of status lights **110** may be illuminated depending upon whether a nurse is talking to the patient, or vice versa.

(38) Headwall unit **94** of FIG. 4 (and headwall unit **94a** of FIG. 5) are adapted to wirelessly receive signals from patient support apparatus **20** and deliver the signals to wall outlet **60** in a manner that matches the way the signals would otherwise be delivered to wall outlet **60** if a conventional nurse call cable (e.g. cable **90**) were connected between patient support apparatus **20** and wall outlet **60**. In other words, patient support apparatus **20** and headwall unit **94** cooperate to provide signals to wall outlet **60** in a manner that is transparent to wall outlet **60** and nurse call system **68** such that these components cannot detect whether they are in communication with patient support apparatus **20** via wired or wireless communication. In this manner, a healthcare facility can utilize the wireless communication abilities of one or more patient support apparatuses **20** without having to make any changes to their existing wall outlet **60** or to their nurse call system **68**.

(39) In addition to sending signals received from patient support apparatus **20** to wall outlet **60**, headwall unit **94** is also adapted to forward signals received from wall outlet **60** to patient support apparatus **20**. Such bidirectional communication includes, but is not limited to, communicating audio signals between a person supported on patient support apparatus **20** and a nurse positioned remotely from patient support apparatus **20** (e.g. nurses' station **72**). The audio signals received by

headwall unit **94** from patient support apparatus **20** are forwarded to wall outlet **60**, and the audio signals received by wall outlet **60** from nurse call system **68** are forwarded to one or more speakers onboard patient support apparatus **20**.

(40) Headwall unit **94** also communicates the data and signals it receives from patient support apparatus **20** to the appropriate pins of wall outlet **60**. Likewise, it communicates the data and signals it receives and/or detects on the pins of wall outlet **60** to patient support apparatus **20** via wireless messages. The wireless messages include sufficient information for patient support apparatus **20** to discern what pins the messages originated from, or sufficient information for patient support apparatus **20** to decipher the information included in the message. In at least one embodiment, headwall unit **94** includes any and/or all of the same functionality as, and/or components of, the headwall units **76** disclosed in commonly assigned U.S. patent application Ser. No. 16/215,911 filed Dec. 11, 2018, by inventors Alexander Bodurka et al. and entitled HOSPITAL HEADWALL COMMUNICATION SYSTEM, the complete disclosure of which is incorporated herein by reference. Alternatively, or additionally, headwall unit **94** may include any and/or all of the same functionality as, and/or components of, the headwall interface **38** disclosed in commonly assigned U.S. patent publication 2016/0038361 published Feb. 11, 2016, entitled PATIENT SUPPORT APPARATUSES WITH WIRELESS HEADWALL COMMUNICATION, and filed by inventors Krishna Bhimavarapu et al., the complete disclosure of which is also incorporated herein by reference. Still further, headwall unit **94** and/or patient support apparatus **20** may include any of the functionality and/or components of the headwall units **140**, **140a** and/or patient support apparatuses **20**, **20a**, and/or **20b** disclosed in commonly assigned U.S. patent application Ser. No. 62/833,943 filed Apr. 15, 2019, by inventors Alexander Bodurka et al. and entitled PATIENT SUPPORT APPARATUSES WITH NURSE CALL AUDIO MANAGEMENT, the complete disclosure of which is incorporated herein by reference.

(41) Cable **90** includes a first end having a first connector **96** and a second end having a second connector **98** (FIG. 6). First connector **96** is adapted to be plugged into wall outlet **60**. Second connector **98** is adapted to be plugged into a cable interface **88** positioned on patient support apparatus **20**. In many healthcare facilities **64**, wall outlet **60** is configured as a 37-pin receptacle. In such facilities, cable **90** includes first and second connectors **96** and **98** having 37 pins (one of which may be a male connector and the other of which may be a female connector, although other combinations may be used). One example of a male 37-pin connector **96**, **98** that may be used as first or second connector **96** or **98** is shown in FIG. 8. One example of a female 37-pin connector **96**, **98** that may be used as first or second connector **96** or **98** is shown in FIG. 9. Other types of 37-pin connectors may also be used, depending upon the configuration of wall outlet **60**. Still further, in some healthcare environments, wall outlet **60** includes fewer pins and/or has an arrangement of pins that is shaped to match a cable **90** having connectors different from what is shown in FIGS. 8 and 9. Patient support apparatus **20** and headwall unit **94** are adapted to communicate with all of these different types of wall outlets **60** via an appropriately selected cable (e.g. one with the proper connectors **96**, **98** on its ends).

(42) FIG. 6 illustrates further details regarding the manner in which patient support apparatus **20** communicates with wall outlet **60**, and vice versa, as well as the structures involved with that communication. In the example of FIG. 6, patient support apparatus **20** has dual communication capabilities. That is, it is able to communicate with wall outlet **60** via either wireless communication link **58** or cable **90**.

(43) Patient support apparatus **20** includes cable interface **88** (FIG. 6), a wireless interface **100**, a controller **112**, a network transceiver **118**, a communication sensor **120**, exit detection system **56**, a battery **122**, a memory **144**, and one or more control panels **48** (only control panel **48a** is shown in FIG. 6, but it will be understood that the functions discussed below with respect to control panel **48a** may be implemented on other ones of the control panels **48b** and/or **48c**).

(44) Cable interface **88** is adapted to electrically couple to the plurality of pins **114** of cable

connector **98**. It will be understood that, although FIG. **6** shows a plurality of pins **114** for each of connectors **96** and **98** of cables **90**, wall outlet **60**, cable interface **88** of patient support apparatus **20**, and a cable interface **116** of headwall unit **94**, one or more of these devices (cable connectors **96**, **98**, cable interfaces **88**, **116**, and wall outlet **60**) will include pin receptacles instead of pins. Such pin receptacles are adapted to receive and electrically couple to pins **114**. Further, it will be understood that it does not matter which of these devices include pins and which includes pin receptacles so long as each connection between the devices include a combination of pins and pin receptacles that allow communication between the mated devices. Consequently, the term “pins” as used herein will refer to pins and/or pin receptacles.

(45) In some embodiments, controller **112** is implemented as, and/or includes, one or more conventional microcontrollers. In other embodiments, controller **112** may be modified to use a variety of other types of circuits-either alone or in combination with one or more microcontrollers-such as, but not limited to, any one or more microprocessors, field programmable gate arrays, systems on a chip, volatile or nonvolatile memory, discrete circuitry, and/or other hardware, software, or firmware that is capable of carrying out the functions described herein, as would be known to one of ordinary skill in the art. Such components can be physically configured in any suitable manner, such as by mounting them to one or more circuit boards, or arranging them in other manners, whether combined into a single unit or distributed across multiple units. The instructions followed by the microcontroller (if included) when carrying out the functions described herein, as well as the data necessary for carrying out these functions, are stored in a memory (e.g. memory **144**) that is accessible to controller **112**.

(46) It will be understood that wall outlet **60**, cables **90**, and cable interfaces **88** and **116** are all illustrated in FIG. **6** as having only six pins. This is done merely for purposes of compact illustration. All of these components typically include 37 pins, although there are other wall outlets having different pin numbers and the principles of the present disclosure can be applied in healthcare facilities having these types of wall outlets as well. The pins that are not shown in FIG. **6** are used by other components of patient support apparatus **20** for other purposes. For example, one or more pins may be used to convey information to a room light **62a** or reading lights **62b** that is electrically coupled to conductor **92**, such as a message or command indicating that the patient has pressed a control on patient support apparatus **20** to turn on or turn off a light (**62a** or **62b**) in the particular room in which patient support apparatus **20** is located. Another pin may communicate the status of a component onboard patient support apparatus **20**, such as, but not limited to whether one or more side rails **34** are in a down position (or an up position); whether the position of any of the side rails **34** changes from an initial state; whether a brake on patient support apparatus **20** is set; whether the exit detection system is armed; whether support deck **30** is at its lowest height; whether head section **40** has pivoted to less than a threshold angle (e.g. 30 degrees); and whether patient support apparatus **20** has been set to monitor a particular set of conditions. These various items of data are detected by one or more corresponding sensors onboard patient support apparatus **20** that are in communication with controller **112**. Still others of the additional pins may be used for still other purposes.

(47) Although FIG. **6** illustrates a number of the pins **114** of cable interface **88** being fed directly to controller **112**, it will be understood that this is done merely for purposes of illustrative convenience, and that one or more of these pins **114** may be fed to one or more intermediary structures before being fed to controller **112**. Such intermediary structures may include, but are not limited to, one or more relays and/or switches whose states are controlled by controller **112**. Additionally, some pins may not be coupled to controller **112**, but may be routed to other structures within patient support apparatus **20**.

(48) It will also be understood that, although FIG. **6** illustrates two cables **90**-one extending from patient support apparatus **20** to outlet **60** and another one extending from headwall unit **94** to a free end, in actual usage within a particular room within a healthcare facility, only a single cable **90** will

be used at any given time. Thus, patient support apparatus **20** will either be coupled to outlet **60** by a cable **90**, or headwall unit **94** will be coupled to outlet **60** by a cable **90**. In the former case, patient support apparatus **20** communicates with wall outlet **60** via wired communication, and in the latter case, patient support apparatus **20** communicates with wall outlet **60** via wireless communication link.

(49) Battery **122** supplies electrical power to patient support apparatus **20** when its power cord (not shown) is not plugged into an external source of electrical energy (e.g. electrical outlet **108** of FIG. **4**). Controller **112**, either alone or in combination with other circuitry, may be adapted to monitor a charge state of battery **122** and/or to monitor one or more other characteristics of battery **122**. When so adapted, controller **112** is adapted to issue an alert if battery **122** drops below a particular charge threshold while patient support apparatus **20** is not coupled to an external source of electrical power.

(50) Controller **112** (FIG. **6**) communicates with communication sensor **120**, exit detection system **56**, control panel **48a**, network transceiver **118**, memory **144**, battery **122**, cable interface **88**, and wireless interface **100**. Controller **112** uses wireless interface **100** to communicate wirelessly with headwall unit **94**, and it uses cable interface **88** to communicate with wall outlet **60** via a cable **90**.

(51) In some embodiments, control panel **48a** includes a display **124** (FIG. **6**) and a plurality of controls **126**. In some embodiments, the display **124** may take on the form and/or functionality of the display **64a** disclosed in commonly assigned U.S. patent application Ser. No. 62/864,638 filed Jun. 21, 2019, by inventors Kurosh Nahavandi et al. and entitled PATIENT SUPPORT APPARATUS WITH CAREGIVER REMINDERS, the complete disclosure of which is incorporated herein by reference. Still other types of displays may also be used. Control panel **48a** may also include a dashboard of the type disclosed in the aforementioned patent application. Still further, patient support apparatus **20** may be configured to issue any of the reminders in any of the manners disclosed in the aforementioned '638 patent application. Other types of control panels **48a**, however, may alternatively be used.

(52) Communication sensor **120** (FIG. **6**) is adapted to detect when cable interface **88** is in communication with wall outlet **60** via a cable **90**. In one embodiment, communication sensor **120** detects the presence of a cable **90** connected between interface **88** and wall outlet **60** by monitoring the voltage, if any, on at least two pins **114a** and **114b** of cable interface **88**. In some embodiments, the two pins are what is commonly referred to as the Nurse Call Plus (+) pin and what is commonly referred to as the Priority Normally Open/Normally Closed (NO/NC) pin. FIG. **10** illustrates a typical arrangement and identification of the pins for a common 37-pin connector. As can be seen therein, pin twenty-five corresponds to the Nurse Call Plus (+) pin, and pin thirty corresponds to the Priority Normally Open/Normally Closed pin. Accordingly, when communication sensor **120** is coupled to a cable interface **88** having the pin arrangement shown in FIG. **10**, communication sensor **120** detects the voltage, if any, on both pins twenty-five and thirty, which correspond to pins **114a** and **114b** of FIG. **6**. This voltage is generated by the nurse call system **68** and delivered to corresponding pins in wall outlet **60** (which in turn are delivered to pins **114a** and **114b** of interface **88** if a cable **90** is connected from wall outlet **60** to patient support apparatus **20**.) If communication sensor **120** detects a voltage on one or both of these pins, it forwards a message to controller **112** indicating that a cable **90** is currently coupled between patient support apparatus **20** and wall outlet **60**. If no voltage is detected on either of these pins, communication sensor **120** forwards a message to controller **112** indicating that no cable **90** is coupled between patient support apparatus **20** and wall outlet **60**.

(53) Communication sensor **120** monitors the voltage on the two pins **114a** and **114b** because it has been found that most manufacturers of nurse call systems **68** will generate a voltage on either or both of these pins of their respective wall outlet **60**. Such voltage typically, although not necessarily always, ranges from about five to twenty-seven volts. Communication sensor **120**, in at least one embodiment, is constructed to detect any voltage that is greater than about 0.3 volts (positive or

negative), although it will be understood that this threshold detection level may be changed. If a voltage of about 0.3 volts or greater is detected on either pin **114a** or pin **114b**, or on both of them, communication sensor **120** concludes that a cable **90** is currently connected between patient support apparatus **20** and wall outlet **60** (and thus nurse call system **68**), and sends a message to controller **112** indicating the existence of this wired connection, as mentioned. If no voltage of about 0.3 volts or greater is detected on either of pins **114a** or **114b**, communication sensor **120** concludes that no wired connection is present and forwards a message indicating such to controller **112**.

(54) In the embodiment shown in FIG. **6**, patient support apparatus **20** is also configured to communicate with wall outlet **60** in a wireless manner (e.g. without cable **90**). This can be seen from the presence of wireless interface **100** on patient support apparatus **20**, and more specifically from the inclusion of radio frequency transceiver **136** within wireless interface **100**. Wireless interface **100** also includes an infrared transceiver **128**. As will be discussed in greater detail below, wireless interface **100** is used to establish a wireless communication link **58** with wall outlet **60** by wirelessly communicating with an adjacent headwall unit **94** that is coupled by a cable **90** to wall outlet **60**.

(55) FIG. **6** illustrates several internal components of headwall unit **94**. These components includes cable interface **116**, a controller **130**, a communication sensor **132**, a battery **134**, an infrared (IR) transceiver **138**, and a Bluetooth transceiver **140**. IR transceiver **138** of wireless headwall unit **94** is adapted to communicate using infrared signals with IR transceiver **128** of patient support apparatus **20**. Bluetooth transceiver **140** of wireless headwall unit **94** is adapted to communicate using Bluetooth communications with Bluetooth transceiver **136** of patient support apparatus **20**.

(56) Controller **130** communicates with communication sensor **132**, transceivers **138** and **140**, as well as with additional electronics that are present on headwall unit **94**. The additional electronics may include any of the electronics disclosed in any of the following commonly assigned patent applications, and wireless headwall unit **94** may be configured to perform any of the functions disclosed in the following commonly assigned patent applications: Ser. No. 16/215,911 filed Dec. 11, 2018, by inventors Alexander Bodurka et al. and entitled HOSPITAL HEADWALL COMMUNICATION SYSTEM; Ser. No. 16/217,203 filed Dec. 12, 2018, by inventor Alexander Bodurka, and entitled SMART HOSPITAL HEADWALL SYSTEM; and Ser. No. 16/193,150 filed Nov. 16, 2018, by inventors Alexander Bodurka et al. and entitled PATIENT SUPPORT APPARATUSES WITH LOCATION/MOVEMENT DETECTION, the complete disclosures of both of which are incorporated herein by reference.

(57) Wireless headwall unit **94** is coupled to a wall outlet **60** by way of cable **90** having a connector **96** that is adapted to be inserted into wall outlet **60**. As was noted, in some embodiments, such as wireless headwall unit **94a** of FIG. **5**, cable **90** may be omitted and wireless headwall unit **94a** may be inserted directly into wall outlet **60** by way of a connector **102**. Further, as was noted previously, although FIG. **6** illustrates both a cable **90** coupling patient support apparatus **20** to wall outlet **60** and a cable **90** extending from headwall unit **94** to a free end, a typical room (or bay of a room) in a healthcare facility will only include a single cable **90** that, depending upon the presence or absence of headwall unit **94**, will extend between patient support apparatus **20** and wall outlet **60**, or between headwall unit **94** and wall outlet **60**. Thus, the two different cables **90** of FIG. **6** illustrate two different options for setting up communications between patient support apparatus **20** and wall outlet **60**.

(58) Cable interface **116** of headwall unit **94** (FIG. **6**) communicates with cable **90** in the same manners as cable interface **88** of patient support apparatus **20**. That is, the signals on the various pins of **114** of wall outlet **60** are communicated to controller **130** via cable **90** and cable interface **116**, and controller **130** is adapted to forward those signals to wireless interface **100** of patient support apparatus **20** using transceivers **138** and/or **140**. Likewise, wireless headwall unit **94** is adapted to receive data from patient support apparatus **20** via one or both of transceivers **138** and/or

140 and to forward the received data, as appropriate, to corresponding pins **114** of connector **98** of cable **90** (which are forwarded to wall outlet **60** when the opposite end (connector **96**) of cable **90** is inserted in wall outlet **60**).

(59) Infrared transceiver **138** of headwall unit **94** acts as a location transceiver. Infrared transceiver **138** is a short range transceiver that emits a short range signal containing an identifier that is unique to that particular wireless headwall unit. Infrared transceiver **128** of patient support apparatus **20** is able to detect the short range signal from infrared transceiver **138** when the patient support apparatus **20** is positioned adjacent to headwall unit **94** (e.g. within approximately a meter or two). Patient support apparatus **20** forwards this unique signal to an off board server, such as server **78** and/or remote server **84**, which contains a table correlating the unique identifiers of each headwall unit **94** to their location within the healthcare facility. This table is generated during a survey of the headwall units **94** when they are initially installed within the healthcare facility. Server **78** is therefore able to determine the location of each patient support apparatus **20** within the healthcare facility whenever the patient support apparatus **20** is positioned adjacent a headwall unit **94**. Further explanation of one manner in which transceivers **128**, **136**, **138**, and **140** may operate are provided in the following commonly assigned U.S. patent applications: Ser. No. 16/215,911 filed Dec. 11, 2018, by inventors Alexander Bodurka et al. and entitled HOSPITAL HEADWALL COMMUNICATION SYSTEM; Ser. No. 16/217,203 filed Dec. 12, 2018, by inventor Alexander Bodurka, and entitled SMART HOSPITAL HEADWALL SYSTEM; and Ser. No. 16/193,150 filed Nov. 16, 2018, by inventors Alexander Bodurka et al. and entitled PATIENT SUPPORT APPARATUSES WITH LOCATION/MOVEMENT DETECTION, the complete disclosures of all of which are incorporated herein by reference.

(60) Bluetooth transceivers **136** and **140** are used by controllers **112** and **130**, respectively, to transmit audio signals between patient support apparatus **20** and wireless headwall unit **94**, such as, but not limited to, the audio signals used to convey the voice signals of the patient and the remotely positioned nurse. Such audio signals may also include the audio signals from television **86** and/or a radio or other entertainment device positioned in the room **70**. Bluetooth transceivers **136** and **140** may also be used to transmit other data, such as, but not limited to, status data regarding the status of patient support apparatus **20**, one or more messages indicating an exit detection alert has been issued, and/or other data. In some embodiments, wireless headwall unit **94** and patient support apparatus **20** are configured to exchange audio signals therebetween in any of the manners disclosed in commonly assigned U.S. patent application Ser. No. 62/833,943 filed Apr. 15, 2019, by inventors Alexander Bodurka et al. and entitled PATIENT SUPPORT APPARATUSES WITH NURSE CALL AUDIO MANAGEMENT, the complete disclosure of which is incorporated herein by reference.

(61) In many embodiments, infrared transceivers **128** and **138** are used to initially establish the Bluetooth communication link between patient support apparatus **20**. In such embodiments, transceivers **128** and **138** may exchange a unique patient support apparatus ID and a unique wireless headwall unit ID. These IDs are then used as addresses for the wireless communication between patient support apparatus **20** and wireless headwall unit **94**. Such addresses ensure that patient support apparatus **20** only establishes a wireless communication link **58** with a headwall unit **94** that is positioned adjacent to the patient support apparatus **20**, even when other wireless headwall units **94** may be within range of Bluetooth transceiver **136**. In other words, patient support apparatus **20** only establishes a wireless communication link **58** with the headwall unit **94** that its IR transceiver **128** has linked to, and due to the IR transceiver's limited range, this only happens when patient support apparatus **20** is positioned adjacent to headwall unit **94**. Still other ways of communicating between patient support apparatus **20** and wireless headwall unit **94** may be utilized.

(62) Communication sensor **132** of wireless headwall unit **94** operates in the same manner as communication sensor **120** of patient support apparatus **20**. That is, communication sensor **132** is

electrically coupled to pins **114a** and **114b** (corresponding to the Nurse Call Plus (+) and Priority Normally Open/Normally Closed (NO/NC) pins) and checks to see if a voltage is detected on either or both of these pins. If it detects a voltage on either or both of these pins, it concludes that connector **98** of cable **90** is coupled to both interface **116** and wall outlet **60**. If it does not detect a voltage on at least one of these pins, it concludes that no cable **90** is coupling interface **116** to wall outlet **60**.

(63) In at least one embodiment, controller **130** of wireless headwall unit **94** is configured to report the output of communication sensor **132** (i.e. whether interface **116** is coupled to wall outlet **60** by a cable **90** or not) to patient support apparatus **20**. That is, controller **130** uses Bluetooth transceiver **140** (or IR transceiver **138**, in some embodiments), to transmit a message to patient support apparatus **20** indicating whether headwall unit **94** is connected to wall outlet **60** or not. In at least one embodiment, patient support apparatus **20** includes an indicator that controller **112** illuminates, displays, or otherwise controls in order to indicate to the user the status of wireless headwall unit **94** vis-a-vis wall outlet **60**. In such embodiments, controller **112** may be configured to illuminate an indicator in a first color (e.g. red or amber) if wireless headwall unit **94** transmits a message to patient support apparatus **20** indicating that it is not coupled to wall outlet **60** (as detected by communication sensor **132**), and to illuminate the indicator in a second color (e.g. green) if wireless headwall unit **94** transmits a message to patient support apparatus **20** indicating that it is coupled to wall outlet **60** (as also detected by communication sensor **132**). In this manner, patient support apparatus **20** provides an indication to the user thereon of the communication status of wireless headwall unit **94**.

(64) In some embodiments of patient support apparatus **20**, it is configured to automatically select a wired or wireless communication method for communicating with wall outlet **60** based on the signals received from communication sensors **120** and **132**. In such embodiments, if communication sensor **120** detects a voltage on pins **114a** and/or **114b**, controller **112** automatically communicates with wall outlet **60** using cable interface **88**. On the other hand, if communication sensor **132** detects a voltage on pins **114a** and/or **114b** and controller **130** sends a message indicating this detection to patient support apparatus **20** (and to controller **112**), controller **112** automatically communicates with wall outlet **60** by sending messages to headwall unit **94** using wireless interface **100**. The communication may include not only the audio signals from the remote nurse and/or patient positioned on patient support apparatus **20**, but also status data regarding patient support apparatus **20**, such as, but not limited to, an alert status of exit detection system **56**, the status of side rails **34** (e.g. raised or lowered), the status of a brake, the height of litter frame **28**, and/or other status data.

(65) In still other embodiments, controller **112** is configured to receive a message from headwall unit **94** indicating whether it should look for cable **90** (via communication sensor **120**). Such a message may be received from headwall unit **94** when headwall unit **94** establishes that a cable **90** is present from itself to wall outlet **60**. Thus, in such cases, controller **112** may disable its monitoring of the presence/absence of a cable that it would otherwise carry out using sensor **120**. As will be discussed, the enablement/disablement of this monitoring feature may be a condition that is synced between patient support apparatus **20** and headwall unit **94** when the two first establish wireless communication link **58**.

(66) In at least one embodiment, controller **112** is configured, after automatically selecting a wired or wireless communication method (e.g. interface **88** or **100**), to continue to use the automatically selected communication method until it either receives a signal from one of the communication sensors **120** or **132** indicating that the currently selected communication method is no longer viable (e.g. a voltage on pins **114a** or **114b** is no longer detected), it is instructed by a caregiver to switch communication methods, or it receives a message from headwall unit **94** indicating what communication method to use (or/or a message to disable/enable the monitoring of cable **90**'s presence). For example, if a caregiver unplugs a cable **90** from wall outlet **60** to patient support

apparatus **20** and thereafter couples a cable **90** between wall outlet **60** and headwall unit **94**, the former act will be detected by communication sensor **120** (if cable monitoring is enabled), while the latter act will be detected by communication sensor **132**. Based on the signals received from these two sensors **120** and **132**, controller **112** may automatically switch from wired communication via interface **88** to wireless communication via interface **100**. Patient support apparatus **20** can therefore automatically select whichever communication method is available without requiring any manual instructions or manipulation of any controls on any of the control panels **48** by the caregiver. However, as noted above,

(67) Network transceiver **118** (FIG. **6**) is a wireless transceiver adapted to communicate with one or more wireless access points **76** of the healthcare facility's local area network **74**. In some embodiments, transceiver **118** may be a WiFi transceiver adapted to transmit and receive wireless electrical signals using any of the various WiFi protocols (IEEE 802.11b, 801.11g, 802.11n, 802.11ac . . . etc.). In other embodiments, network transceiver **118** may be a transceiver adapted to communicate using any of the frequencies, protocols, and/or standards disclosed in commonly assigned U.S. patent application Ser. No. 62/430,500 filed Dec. 6, 2016, by inventor Michael Hayes and entitled NETWORK COMMUNICATION FOR PATIENT SUPPORT APPARATUSES, the complete disclosure of which is incorporated herein by reference. In still other embodiments, transceiver **118** may be a wired transceiver that communicates with network **74** over a wired network, such as an Ethernet cable or the like. Regardless of whether transceiver **118** is a wired or wireless transceiver, it enables controller **112** to communicate with one or more servers on the healthcare facility's network **72**, such as, but not limited to, patient support apparatus server **78**.

(68) Controller **112** uses network transceiver **118** to send messages to server **78** (and/or server **84**) indicating the status patient support apparatus **20**, headwall unit **94**, and/or other information. Such status information includes data indicating whether patient support apparatus **20** is communicating via cable interface **88** or wireless interface **100**, whether cable **90** is coupled between interface **88** and outlet **60**, whether a cable is coupled between interface **116** and outlet **60**, well as status data regarding battery **122** and/or **134**, and other information. In some embodiments, server **78** and/or server **84** are configured to share this data with one or more other devices within the healthcare facility. For example, in at least one embodiment, server **78** and/or server **84** are configured to transmit the communication status of patient support apparatus **20** to one or more electronic devices **142** (FIG. **4**), such as the electronic devices **104a** and/or **104b** disclosed in commonly assigned U.S. patent application Ser. No. 62/868,947 filed Jun. 30, 2019, by inventors Thomas Durlach et al. and entitled CAREGIVER ASSISTANCE SYSTEM, the complete disclosure of which is incorporated herein by reference. The electronic devices **142** may be smart phones, tablet computers, laptop computers, desktop computers, and/or other computing devices that are in communication with network **74** and therefore able to receive data from patient support apparatus server **78**.

(69) Memory **144** may be comprised of any one or more of the following: non-volatile flash memory, Random Access Memory (RAM), Read Only Memory (ROM), a mechanical hard drive, a solid state hard drive, etc. Memory **144** may contain not only the instructions followed by controller **112** and the data used for carrying out those instructions, but also data defining the current status of one or more conditions, some of which may pertain to components onboard patient support apparatus **20** (e.g. exit detection system **56**, siderails **354**, etc.), and some of which may pertain to components off-board patient support apparatus **20** (e.g. room light **62a**, reading light **62b**, TV **86**, nurse call system **68**, and/or headwall unit **94**). As will be discussed in greater detail below, the state values of some of these conditions are automatically synced with the state values of corresponding conditions that are present onboard headwall unit **94** when patient support apparatus **20** and headwall unit **94** first establish communications with each other.

(70) FIG. **7** illustrates one example of a patient control panel **48c** that may be incorporated into patient support apparatus **20** and positioned at a location on patient support apparatus **20** that is convenient for a patient to access while supported on support deck **30**, such as on an interior side of

one of the siderails **34**. Control panel **48c** includes a plurality of controls **126a-n** that are intended to be operated by a patient. A nurse call control **126a**, when pressed by the patient, sends a signal to the nurse call system **68** requesting that a remotely positioned nurse talk to the patient. A Fowler-up control **126b**, when pressed by the patient, causes a motorized actuator onboard patient support apparatus **20** to raise Fowler section **40** upwardly. A Fowler-down control **126d**, when pressed by the patient, causes the motorized actuator to lower Fowler section **40** downwardly. A gatch-up control **126c**, when pressed by the patient, causes another motorized actuator to raise a knee section of support deck **30**, while a gatch-down control **126e** causes the motorized actuator to lower the knee section of support deck **30**.

(71) A volume-up control **126f**, when pressed by the patient, causes patient support apparatus **20** to send a signal to an in-room television **86** instructing it to increase its volume, while a volume down control **126h**, when pressed, causes patient support apparatus **20** to send a signal to the television **86** instructing it to decrease its volume. A channel-up control **126g**, when pressed by the patient, causes patient support apparatus **20** to send a signal to the television **86** instructing it to increase the channel number, while a channel-down control **126i**, when pressed, causes patient support apparatus **20** to send a signal to the television instructing it to decrease the channel number.

(72) A mute control **126j**, when pressed, causes patient support apparatus **20** to send a signal to the television instructing it to either mute itself or unmute itself, depending upon whether the television is currently muted or unmuted. In other words, mute control **126j** is a toggle control that alternately sends mute and unmute commands to the television when it is pressed.

(73) Power control **126k** (FIG. 7) is a toggle control that, when pressed, sends a signal to the television to either turn on or turn off, depending upon the television's current power status. Closed-captioning control **126l** is another toggle control that, when pressed, sends a signal to the television to either turn on its closed-captioning feature or to turn off its closed captioning feature, depending upon whether the closed-captioning feature is currently on or off.

(74) Control **126m** is a toggle control that, when pressed, sends a signal to a room light **62a** to either turn on or turn off, depending upon the current state of the room light **62a**. Control **126n** is another toggle control that, when pressed, sends a signal to reading light **62b** to either turn on or turn off, depending upon the current state of the reading lights **62b**. Both of lights **62a** and **62b** are positioned off-board the patient support apparatus **20**.

(75) It will be understood that not only the number of controls **126** on control panel **48c** (FIG. 7), but also the functions of the controls **126** on control panel **48c**, the layout of the controls **126** on control panel **48c**, and/or other aspects of control panel **48c** may be modified from what is shown in FIG. 7. In some embodiments, control panel **48c** is implemented on a pendant controller that includes a cable that is plugged into a port on patient support apparatus **20**. Still other manners of implementing control panel **48c** are also possible.

(76) Patient support apparatus **20**, in some embodiments, communicates various alerts and data through cable interface **88** to wall outlet **60** by changing the state of one or more relays, or their electrical equivalent, that are in communication with different pairs of the pins **114**. Thus, for example, if an exit alert is detected by exit detection system **56**, controller **112** may change a state of a first relay from being open to being closed, or vice versa, wherein that first relay is in electrical communication with a pair of pins **114** in cable interface **88**. Similarly, if a patient presses a nurse call control (e.g. control **126a**) indicating his/her request to speak with a remotely positioned nurse via nurse call system **68**, controller **112** may change the state of another relay (from open to closed, or vice versa) that is in electrical communication with another pair of pins **114** in cable interface **88**. The same opening or closing of a corresponding relay may also occur in response to the patient activating control **126m** (FIG. 7) on patient support apparatus **20** for turning on/off a room light **62a**, the patient activating control **126n** on patient support apparatus **20** for turning on/off a reading light **62b**, and/or the patient activating a control **126k** on patient support apparatus **20** for turning on/off a television **86**.

(77) Headwall unit **94** (FIG. **6**) may also contain a plurality of relays (not shown), or their electrical equivalent, that are in electrical communication with pairs of pins **114** in cable interface **116**. During normal operation of headwall unit **94** (i.e. after the initial syncing process described herein is completed), controller **130** controls the states of these relays based on messages received from patient support apparatus **20** via IR transceiver **138** and/or RF transceiver **140**. Thus, for example, if patient support apparatus **20** detects a patient exit and is configured to communicate that by closing a first relay that electrically couples together pins **30** and **31** (see FIG. **11**) of cable interface **88**, it will send an exit detection alert message to headwall unit **94** via one or more both of transceivers **138**, **140**, and controller **130** of headwall unit **94** will react by closing a first relay coupled to cable interface **116** that electrically couples together the same pair of pins **30** and **31**.

(78) A similar process happens for other events that occur on patient support apparatus **20**, such as a nurse call being requested by the patient (control **126a**), the reading light being turned on/off (control **126n**), the room light being turned on/off (**126m**), and/or the TV being turned on/off (control **126k**). That is, whenever one of these actions occurs on patient support apparatus **20**, controller **112** sends a corresponding message to headwall unit **94** that causes controller **130** to open/close a relay inside headwall unit **94**. The relay that is opened/closed by controller **130** changes the electrical state of a pair of pins in cable interface **116**, and the particular pair of pins whose electrical state is changed by this opening/closing of a relay are the same pair of pins whose electrical state controller **112** changes in cable interface **88**.

(79) In addition to changing the states of various relays, controller **130** of headwall unit **94** also monitors the states (i.e. voltages) on various pins **114** of cable interface **116** and sends messages to patient support apparatus **20** via transceiver(s) **140**, **138** that inform patient support apparatus **20** of the states of these pins. Thus, for example, headwall unit controller **130** monitors a first pin **114** (or pair of pins **114**) that is/are associated with a nurse call light. The nurse call light, in some embodiments, controls a backlight for control **126a** that, when activated, changes a state of a backlight positioned behind control **126a**. Alternatively, or additionally, the nurse call light may be positioned elsewhere on control panel **48c** (FIG. **7**), or at other locations on patient support apparatus **20**.

(80) In some embodiments, the voltage across pins **25** and **28** (FIG. **11**) is associated with the state of the nurse call light, and controller **130** monitors this voltage to see what state the nurse call system **68** is indicating that the nurse call light should be in (on/off). The nurse call light is illuminated (or has its illumination state, such as its color, changed) when the patient presses nurse call control **126a** to request to speak with a remotely positioned nurse, and the nurse call system **68** acknowledges receipt of this call. Notification of the activation of this control is sent to the nurse call system **68** via the opening/closing of a relay coupled to specific pins of wall outlet **60** (as discussed above), and the nurse call system **68**, in response to successfully receiving this request, sends a signal back to the patient support apparatus **20** indicating that it should illuminate the nurse call light (or otherwise change its illumination state) as an acknowledgement by the nurse call system **68** that a call has been successfully placed. This response is communicated by changing the voltage across a pair of pins (e.g. pins **25** and **28** of FIG. **11**). When controller **130** of headwall unit **94** detects a voltage across the nurse call request pins (**25** and **28** of FIG. **11**) of cable interface **116**, it sends a message to patient support apparatus **20** (via transceivers **138** and/or **140**) that causes controller **112** of patient support apparatus **20** to illuminate the nurse call light (or otherwise change its illumination state).

(81) Controller **130** of headwall unit **94** (FIG. **6**) also monitors the voltage between other pairs of pins **114** of cable interface **116** and reports, as appropriate, the state of this voltage to patient support apparatus **20** via one or both of transceivers **138**, **140**. One such other pair of pins are pins **114** associated with a nurse call answer light (see, e.g. pins **16** and **29** of FIG. **11**). A nurse call answer light is a light that the nurse call system **68** instructs the patient support apparatus **20** to illuminate (or have its illumination state changed) when a nurse actually answers the patient's call.

Thus, the nurse call answer light differs from the nurse call light in that the former is illuminated when the nurse answers a call while the latter is illuminated when the nurse call system **68** acknowledges that the call was placed.

(82) When patient support apparatus **20** is coupled to wall outlet **60** via a cable **90**, controller **112** of patient support apparatus **20** performs the same type of monitoring of the pins **114** of cable interface **88** that controller **130** does with respect to cable interface **116**. In such situations, controller **112** takes the same actions that it does when headwall unit **94** sends it a message informing it of a change in the state of a particular pair of pins. For example, controller **112** of patient support apparatus **20** may monitor the voltage across pins **24** and **28** (FIG. **11**) to determine whether or not to illuminate its nurse call light, as well as to monitor the voltage across pins **16** and **29** to determine whether or not to illuminate its nurse call answer light. Controller **112** reacts to changes in voltage across these pins in the same manner as it reacts to a message from headwall unit **94** informing it of a similar change detected in the corresponding pins of cable interface **116**. That is, controller **112** controls the illumination state of the nurse call light and/or the nurse call answer light in response to changes in the voltage across these pins.

(83) It can therefore be seen that headwall unit **94** controls the voltages across various pins **114** in cable interface **116** (when patient support apparatus **20** is communicating wirelessly with wall outlet) in the same manner that patient support apparatus **20** controls the voltage across the corresponding pins **114** in cable interface **88** (when patient support apparatus **20** is communicating with wall outlet **60** via a cable **90**). Similarly, headwall unit **94** monitors the voltage across various other pins **114** of cable interface **116** and sends messages to patient support apparatus **20** that cause patient support apparatus **20** to take the same actions that it would take in response to controller **112**'s detection of those same voltage changes on cable interface **88**. Thus, cable interface **116** acts as a proxy for cable interface **88** when no cable is connected between patient support apparatus **20** and wall outlet **60**.

(84) FIG. **8** illustrates a table **150** in which are listed a plurality of conditions **152**, a syncing direction **154**, and examples of types of values **156** for the conditions **152**. The conditions **152** identified in FIG. **8** correspond to various states that headwall unit **94** and/or patient support apparatus **20** may be in when they are initially paired together via wireless communication link **58**. Because each device (patient support apparatus **20** and headwall unit **94**) has its own copy of its current states for these conditions **152**, and because the copy that each device has may not match the copy that the other device has when the devices are initially paired, both patient support apparatus **20** and headwall unit **94** are configured to executed a syncing process when they are initially paired. The syncing process matches a first subset of the conditions values that are present on the patient support apparatus **20** to those that are present on headwall unit **94** when they are first paired together, and matches other subset of the condition values that are present on headwall unit **94** to those that are present on headwall unit **94** when they are first paired together. By matching the values of some of patient support apparatus **20**'s conditions to headwall unit **94**'s condition values, and by matching the values of some of headwall unit **94**'s conditions to patient support apparatus **20**'s condition values, the syncing process ensures that, after initially being paired, both patient support apparatus **20** and headwall unit **94** are in agreement as to the current values for all of the conditions. Examples of several of these conditions and their various states will now be described in greater detail.

(85) Some of the conditions that are synced between patient support apparatus **20** and headwall unit **94** when they first establish communications are the states of the relays present in each device (some of which were discussed above), while other conditions that are synced may refer to states that are not associated with relays. In either case, the current values for each condition may be stored in memories on each device.

(86) During the initial syncing process, headwall unit **94** may have its relays in a first state while patient support apparatus **20** may have its relays in another state. When the headwall unit **94** and

patient support apparatus **20** are first paired together, the question arises as to which relay state (that of the patient support apparatus's or that of the headwall unit's) is going to take precedence over the other. In other words, does patient support apparatus **20** change its relay states to match those of headwall unit **94**, or does headwall unit **94** change its relay states to match those of patient support apparatus **20**? Table **150** of FIG. **8** provides the answer to this question for each relay (as well as for multiple non-relay conditions that are synced between patient support apparatus **20** and headwall unit **94** in response to communication link **58** being established).

(87) Specifically, column **154**—the sync direction—indicates whether the patient support apparatus's state or the headwall unit's state will take priority over the other's state when they differ (if they don't differ, then the question of which one takes priority is moot). For each row in table **8** that states “bed_to_wall,” this means that the patient support apparatus's current state of the condition listed in that row takes precedence over the headwall unit's current state of that condition when the two devices are synced together. Conversely, for each row in table **8** that states “wall_to_bed,” this means that the headwall unit's current state of the condition listed in that row takes precedence over the patient support apparatus's current state of that condition when the two devices are synced together. This may be better understood with respect to several examples, which will now be described.

(88) When patient support apparatus **20** and headwall unit **94** are first paired together, the relay inside of patient support apparatus **20** that controls the reading light **62b** (condition **152a** of FIG. **8**) may be in a different state than the relay inside of headwall unit **94** that controls the reading light **62b**. Thus, headwall unit **94** may, for example, be in a state where the reading light condition **152a** has an off value (i.e. reading light **62b** is turned off) and patient support apparatus **20** may be in a state where the reading light condition **152** has an on value (i.e. reading light **62b** is turned on). When patient support apparatus **20** and headwall unit **94** are synced together, the state of reading lights **62b** as set in patient support apparatus **20** takes precedence over the state of reading light **62b** as set in headwall unit **94** (as indicated by column **154** of table **150** (FIG. **8**). Thus, patient support apparatus **20** sends one or more messages to headwall unit **94** during the syncing process indicating its value of reading light condition **152a**, and controller **130** of headwall unit **94** changes its value of reading light condition **152a** to match that of the patient support apparatus's.

(89) Similarly, if the states of the room light **62a** (condition **152b** of FIG. **8**) differ between patient support apparatus **20** and headwall unit **94** when the two are initially paired together, the state of this condition **152b** in patient support apparatus **20** takes precedence over the state of this condition in headwall unit **94**. Patient support apparatus **20** sends one or more messages to headwall unit **94** during the syncing process indicating its current state of room light condition **152b**, and controller **130** of headwall unit **94** changes its state of room light condition **152b** to match that of the patient support apparatus's. In some embodiments, the state values of conditions **152a** and **152b** may correspond to voltages and/or open or closed states of relays associated with pins **2** and **3** (FIG. **11**), although the values of these conditions **152a** and/or **152b** may be associated with other and/or additional pins **114**.

(90) As shown in FIG. **8**, the state of conditions **152e**, **152f**, **152g**, **152s**, and **152v** within patient support apparatus **20** all take precedence over the corresponding states of these conditions within headwall unit **94** during the syncing process. Controller **112** of patient support apparatus **20**, as part of the syncing process, sends the current values for the states of these conditions to controller **130** of headwall unit **94**, and controller **130** changes its corresponding state values for these conditions to match those received in the message(s) from patient support apparatus **20**.

(91) Condition **152e** (FIG. **8**) refers to the condition of nurse call request control **126a**. If a patient has activated the nurse call control **126a**, then the state value of condition **152e** is set by patient support apparatus **20** to “on.” If the patient has not activated the nurse call control **126a**, then the state value of condition **152e** is set by patient support apparatus **20** to “off.” As was discussed previously, controller **112** may set the state of condition **152e** to these two different values by

controlling a relay associated with a specific pair of pins **114** of cable interface **88** (e.g. pins **25** and **26**, FIG. **11**) and/or by storing a corresponding value in memory **144**.

(92) Condition **152f** (FIG. **8**) refers to the condition of an exit detection alert (or other high priority alert) that takes place on patient support apparatus **20**. If exit detection system **56** has detected a patient exit, then controller **112** sets the state value of condition **152f** to “on.” If exit detection system **56** has not detected a patient exit, then controller **112** sets the state value of condition **152f** to “off.” As was discussed previously, controller **112** may set the state of condition **152f** to these two different values by controlling a relay associated with one or more pairs of pins **114** of cable interface **88** (e.g. pins **30** and **31**, FIG. **11**), and/or by storing a corresponding value in memory **144**.

(93) Condition **152g** (FIG. **8**) refers to the condition of television **86**. If a patient has utilized control **126k** to turn on television **86**, then controller **112** sets the state value of condition **152g** to “on.” If the patient has utilized control **126k** to turn off television **86** (or the patient has not utilized control **126k** to turn on television **86**), then controller **112** sets the state value of condition **152g** to “off.” As was discussed previously, controller **112** is able to control television **86** to match condition **152g** by controlling a relay and/or voltage associated with pins **33** and/or **34** (FIG. **11**).

(94) Condition **152s** and **152v** (FIG. **8**) refers to conditions of patient support apparatus **20** that may not be associated with particular pins **114**. Condition **152s** refers to a request from patient support apparatus **20** to headwall unit **94** for a diagnostics report from headwall unit **94**. If patient support apparatus **20** has this condition recorded in its memory **144** as being requested, it sets the value of this condition to “yes.” If patient support apparatus **20** has this condition recorded in its memory **144** as not being requested, it sets the value of this condition to “no.”

(95) Condition **152v** refers to the condition of an IR session ID. Whenever patient support apparatus **20** and headwall unit **94** are paired together, IR transceiver **128** and IR transceiver **138** assign an identifier to their communication session. Patient support apparatus **20** records this session ID in its memory and forwards it to headwall unit **94** during the syncing process. When controller **130** of headwall unit **94** receives it, it overwrites, or otherwise syncs, its IR session ID value with the one received from patient support apparatus **20** during the syncing process.

(96) Headwall unit **94** is configured to have its values of all of the remaining conditions **152** shown in table **150** of FIG. **8** take precedence over the corresponding values of these conditions in patient support apparatus **20**. That is, headwall unit **94**'s values of conditions **152c**, **152d**, **152h**, **152i**, **152j**, **152k**, **152l**, **152m**, **152n**, **152o**, **152p**, **152q**, **152r**, **152t**, and **152u** all take precedence over patient support apparatus **20**'s values of these conditions during the syncing process. Each of these conditions will now be described in additional detail.

(97) Condition **152c** (FIG. **8**) refers to the condition of the nurse answer light discussed above. If a remotely positioned nurse has answered a patient's call placed to the nurse call system **68**, then condition **152c** is in an on state. If the nurse call system **68** has indicated that a nurse has not answered a patient's call (or no such patient call has been placed), then condition **152c** is in an off state. As was discussed previously, controller **130** of headwall unit **94** may determine the state of the nurse answer light by determining the state of one or more specific pins **114** of cable interface **116** (e.g. pins **16** and **29**).

(98) Condition **152d** (FIG. **8**) refers to the condition of the nurse call light discussed above. If the nurse call system **68** has acknowledged a patient call placed from patient support apparatus **20** (via control **126a**), then condition **152d** is in an on state. If the nurse call system **68** has not acknowledged a patient call (or no such patient call has been placed), then condition **152d** is in an off state. As was discussed previously, controller **130** of headwall unit **94** may determine the state of the nurse call light by determining the state of one or more specific pins **114** of cable interface **116** (e.g. pins **25** and **28**).

(99) Conditions **152h-r** and **152t-u** (FIG. **8**) refers to conditions of patient support apparatus **20** that may not be associated with particular pins **114**. Values for these conditions **152** (as well as the previously discussed conditions **152**) may be stored in a memory onboard patient support apparatus

20 (e.g. memory **144**), as well as a memory onboard headwall unit **94** (not shown).

(100) Condition **152h** refers to a type of TV **86** that is positioned within the same room as headwall unit **94**. The type, in some embodiments, refers to a brand of television, and patient support apparatus **20** and/or headwall unit **94** have built-in intelligence indicating the sequence of voltages, and/or other messages, necessary to send different commands to different types of televisions in order to effectuate the desired response. In other words, for example, some TV brands require a certain sequence of voltages to raise the channel number, while other TV brands may require a different sequence of voltages (and/or different voltage levels) to raise the channel number. The state value for condition **152h** identifies the brand/type of TV **86** associated with headwall unit **94** so that the TV **86** can be properly controlled.

(101) Condition **152i** (FIG. **8**) refers to the open or closed (e.g. on or off) state of a pair of pins **114** that are to be interlocked together (i.e. electrically shorted together) in order for certain brands and/or types of nurse call systems **68** coupled to wall outlet **60** to be able to detect the absence/presence of patient support apparatus **20**. Condition **152j** refers to the open or closed (e.g. on or off) state of another pair of pins **114** that are to be interlocked together for other (sometimes referred to as “standard”) nurse call systems **68** coupled to wall outlet **60** to be able to detect the absence/presence of patient support apparatus **20**. Condition **152k** refers to the open or closed (e.g. on or off) state of another pair of pins **114** that are to be interlocked together in order for audio signals to be properly communicated to/from the nurse call system **68** to wall outlet **60**. For some nurse call systems **68**, the absence of a closed circuit between one or more pairs of these interlock pins causes the nurse call system **68** to issue a “cord-out” alert, which is a signal to one or more nurses that the patient support apparatus **20** and/or its cable **90** has become disconnected from wall outlet **60**.

(102) Condition **152l** refers to an interlock forcing function (that may be on or off) wherein the patient support apparatus **20** and/or headwall unit **94** couple together one or more of the interlock pins **114** of conditions **152i**, **j**, and/or **k** without waiting to detect any voltage between certain pins of outlet **60**. In other words, when this function is turned on, patient support apparatus **20** or headwall unit **94** (whichever one is coupled to wall outlet **60** by a cable) will electrically short one or more of the interlock pins **114** without waiting to detect any signals from the nurse call system **68**. When this function is turned off, patient support apparatus **20** or headwall unit **94** will not electrically short one or more of the interlock pins **114** until it detects certain voltages on one or more pins **114**.

(103) Condition **152m** refers to the on/off state of a cable detection function carried out by patient support apparatus **20**. As was discussed above, controller **112** of patient support apparatus **20** is configured, when condition **152m** is in an on state, to monitor the outputs of communication sensor **120** to detect whether a cable **90** is plugged into interface **88**, and to switch to using cable interface **88** to communicate with wall outlet **60** when a cable **90** is detected. When condition **152m** is in an off state, controller **112** stops monitoring the communication sensor **120**'s outputs and/or does not automatically switch over to using cable interface **88** for communication with wall outlet **60** if a cable **90** is detected.

(104) Condition **152n** refers to the amplification level (e.g. dB) that headwall unit **94** applies to the audio signals it receives from wall outlet **60** (e.g. a nurse's voice signals from nurse call system **68**) and/or the audio signals it transmits to nurse call system **68** (e.g. the patient's voice signals). Condition **152o** refers to the type (e.g. brand) of nurse call system **68** that is coupled to the particular wall unit **60** in that particular room.

(105) Condition **152p** refers to the voltage levels that the nurse call system **68** applies to various of the pins **114** of wall outlet **60**, and/or the voltage levels that headwall unit **94** and/or patient support apparatus **20** have to supply to cable interfaces **116**, **88** in order for the nurse call system **68** to properly interpret the data being communicated to nurse call system **68**.

(106) Condition **152q** refers to whether cable **90** is currently connected from headwall unit **94** to

wall outlet **60** and, as was discussed above, may be determined by communication sensor **120**. Condition **152r** refers to whether or not headwall unit **94** has been configured yet or not. Headwall units **94**, when initially installed within a particular healthcare facility, are configured to match the healthcare facility's usage of the different pins in wall outlet **60**, as well as to know the correct voltage levels and/or other signals that are transmitted to wall outlet **60** so that proper communication can take place between patient support apparatus **20**, nurse call system **68**, TV **86**, and lights **62a** and **62b**.

(107) Condition **152t** refers to whether or not headwall unit **94** has been synced with the currently paired patient support apparatus **20**. And condition **152u** refers to the current charge status of battery **134**, which may be expressed with different values (e.g. a percentage of charge remaining; needs/doesn't need replacement; hi, medium, or low charge, etc.).

(108) The syncing process carried out by patient support apparatus **20** and headwall unit **94** starts, in some embodiments, immediately after the two devices are paired together. The syncing process begins by one of the two devices (**20** or **94**) sending to the other device its current values of a first subset of the conditions **152** shown in FIG. **8**. The recipient device then syncs its values of those conditions to the values that it received from the sending device. The recipient device then sends the original sending device its current values of a second and different subset of the conditions **152** shown in FIG. **8**. The recipient of that message (i.e. the original sending device) then syncs its values of those conditions to the values that it received from the original recipient device. The devices are then synchronized.

(109) The subsets of the values that are sent refer to those conditions of which the sending device's values take precedence over the receiving device's values. In other words, with respect to FIG. **8**, the subset of values that patient support apparatus **20** sends to headwall unit **94** for syncing are those values associated with conditions **152a**, **152b**, **152e**, **152f**, **152g**, **152s**, and **152v**. Patient support apparatus **20** sends these values to headwall unit **94** and headwall unit **94** syncs its values of these conditions so that they match what patient support apparatus **20** sent.

(110) The subset of values that headwall unit **94** sends to patient support apparatus **20** for syncing are those values associated with the rest of the conditions shown in table **150** of FIG. **8**. Specifically, headwall unit **94** sends its values of conditions **152c-d**, **152h-r**, and **152t-u** to patient support apparatus **20** during the syncing process and controller **112** of patient support apparatus **20** syncs its values of these conditions so that they match what headwall unit **94** sent. After each device has transmitted its subset of condition values to the other device, and both devices have indicated to each other that they have synced their condition values with the subset of values they received from the other device, the syncing process is completed.

(111) The syncing process thereafter doesn't occur again until the communication link between patient support apparatus **20** and headwall unit **94** is terminated, and these two devices are paired together again. Thus, when a patient support apparatus **20** is moved from a first room to a second room, it will automatically sync itself with the first headwall interface **94** it pairs with in the first room (and that headwall unit **94** will also automatically sync itself with the patient support apparatus **20**), and subsequently the patient support apparatus **20** will automatically sync itself with a second headwall interface **94** in the second room (which will also sync itself to the patient support apparatus **20** when the two pair together) when the patient support apparatus **20** is moved to that new room. Still further, if the patient support apparatus **20** is moved from the first headwall unit **94** to a different room that has no headwall unit **94**, but instead has only a cable **90** coupled to wall outlet **60**, the patient support apparatus **20** will use the last set of conditions values that it possessed when communicating with the headwall unit **94** for communicating with the wall outlet via cable **90**.

(112) It will be understood that various modification may be made to the syncing process disclosed herein. These modifications include, but are not limited to, syncing fewer conditions **152** than what is shown in FIG. **8**, syncing more conditions **152** than what is shown in FIG. **8**, and/or syncing

different conditions 152 than what is shown in FIG. 8. In some embodiments, patient support apparatus 20 is further configured to store additional information about the current settings and/or preferences of the patient assigned to patient support apparatus 20. For example, in some embodiments, controller 112 of patient support apparatus 20 may record the current channel that a patient is watching on television 86 in a first room and then, if the patient support apparatus 20 is moved to a different room with a different television 86, automatically send a message to the different television 86 in the different room to turn to the channel the patient was watching in the previous room. Controller 112 of patient support apparatus 20, may also, or alternatively, be configured to store other patient preferences, such as channel volume, closed captioning, and/or other preferences, and automatically transfer them to a television 86 in a different room so that the television is automatically synced with the patient's preferences when the patient support apparatus 20 is moved to that room.

(113) It will be understood by those skilled in the art that the use of the term “transceiver” throughout this specification is not intended to be limited to devices in which a transmitter and receiver are necessarily within the same housing, or share some circuitry. Instead, the term “transceiver” is used broadly herein to refer to both structures in which circuitry is shared between the transmitter and receiver, and transmitter-receivers in which the transmitter and receiver do not share circuitry and/or a common housing. Thus, the term “transceiver” refers to any device having a transmitter component and a receiver component, regardless of whether the two components are a common entity, separate entities, or have some overlap in their structures.

(114) Various additional alterations and changes beyond those already mentioned herein can be made to the above-described embodiments. This disclosure is presented for illustrative purposes and should not be interpreted as an exhaustive description of all embodiments or to limit the scope of the claims to the specific elements illustrated or described in connection with these embodiments. For example, and without limitation, any individual element(s) of the described embodiments may be replaced by alternative elements that provide substantially similar functionality or otherwise provide adequate operation. This includes, for example, presently known alternative elements, such as those that might be currently known to one skilled in the art, and alternative elements that may be developed in the future, such as those that one skilled in the art might, upon development, recognize as an alternative. Any reference to claim elements in the singular, for example, using the articles “a,” “an,” “the” or “said,” is not to be construed as limiting the element to the singular.

Claims

1. A headwall unit adapted to be mounted to a headwall of a room in a healthcare facility, the headwall unit comprising: a nurse call interface adapted to electrically couple to a wall outlet mounted in the headwall, the nurse call interface including a set of pins adapted to electrically couple to a plurality of conductors defined in the wall outlet when the nurse call interface is coupled to the wall outlet, each of the pins adapted to be in different electrical states; a first wireless transceiver adapted to wirelessly communicate with a second wireless transceiver positioned onboard a patient support apparatus when the patient support apparatus is positioned adjacent to the headwall unit; a controller adapted to use the first wireless transceiver to establish a communication link with the patient support apparatus, the controller further adapted to perform a syncing process in response to the communication link being initially established, the syncing process comprising: receiving a plurality of state values from the patient support apparatus, the plurality of state values corresponding to a current state of the patient support apparatus; and syncing the electrical states of a subset of the set of pins to match the state values received from the patient support apparatus; wherein the controller is further adapted, after completing the syncing process, to change an electrical state of at least one pin in the subset of the set of pins in response to

receipt of a message from the patient support apparatus indicating a change to the current state of the patient support apparatus.

2. The headwall unit of claim 1 wherein a first one of the state values received from the patient support apparatus defines an on or off state of a reading light, the reading light being adapted to provide illumination to a patient on the patient support apparatus, and wherein a second one of the state values received from the patient support apparatus defines an on or off state of a room light, the room light being adapted to provide illumination to an entire room in which the headwall unit is positioned.

3. The headwall unit of claim 1 wherein a first one of the state values received from the patient support apparatus defines an on or off state of a nurse call request, the nurse call request being adapted to notify a nurse call system coupled to the wall outlet of a patient's desire to speak with a nurse.

4. The headwall unit of claim 1 wherein a first one of the state values received from the patient support apparatus defines an on or off state of a priority alert signal, the priority alert signal being adapted to notify a nurse call system of a priority event occurring on the patient support apparatus.

5. The headwall unit of claim 1 wherein a first one of the state values received from the patient support apparatus defines an on or off state of a television control signal, the television control signal being adapted to instruct a television to be turned on or off.

6. The headwall unit of claim 1 further comprising a third wireless transceiver adapted to wirelessly communicate with a fourth wireless transceiver positioned onboard the patient support apparatus when the patient support apparatus is positioned adjacent to the headwall unit, wherein the communication link is established when both the first wireless transceiver and the third wireless transceiver are in communication with the second and fourth wireless transceivers, respectively, onboard the patient support apparatus, and wherein the first wireless transceiver is a radio frequency transceiver and the third wireless transceiver is an infrared transceiver.

7. The headwall unit of claim 1 wherein the controller is further adapted to send a second plurality of state values to the patient support apparatus in response to the communication link being established, the second plurality of state values corresponding to a current electrical state of a second subset of the set of pins, the second subset different from the subset.

8. The headwall unit of claim 1 wherein the controller is further adapted to receive audio signals from the patient support apparatus via the first wireless transceiver, to forward the audio signals to at least one of the pins, to receive a configuration message via the first wireless transceiver, wherein the configuration message includes an identification of a first pair of the pins that must be electrically shorted together to avoid triggering a cord-out alert in a nurse call system coupled to the wall outlet, and to receive a new configuration message via the first wireless transceiver, wherein the new configuration message includes a new identification of a second pair of pins that must be electrically shorted together to avoid triggering the cord-out alert, wherein the second pair of pins is different from the first pair of pins.

9. The headwall unit of claim 1 wherein the controller is further adapted to receive an exit alert message from the patient support apparatus and to change an electrical state of at least one of the pins in response to receiving the exit alert message.

10. The headwall unit of claim 1 further comprising a memory in which is stored a unique identifier that uniquely identifies the headwall unit, wherein the controller is further adapted to forward the unique identifier to the patient support apparatus using the first wireless transceiver.

11. A patient support apparatus comprising: a support surface adapted to support a patient thereon; a first wireless transceiver adapted to wirelessly communicate with a second wireless transceiver positioned onboard a headwall unit mounted to a headwall of room in a healthcare facility; a controller adapted to use the first wireless transceiver to establish a communication link with the headwall unit, the controller further adapted to perform a syncing process in response to the communication link being initially established, the syncing process comprising: receiving a

plurality of headwall state values from the headwall unit, the plurality of headwall state value corresponding to a current state of the headwall unit; and syncing a plurality of patient support apparatus state values to match the headwall state values received from the headwall unit; wherein the controller is further adapted, after completing the syncing process and in response to a change in a current state of the patient support apparatus, to transmit a message to the headwall unit instructing the headwall unit to change one of the plurality of headwall state values.

12. The patient support apparatus of claim 11 further comprising a nurse answer light adapted to be illuminated when a remotely positioned nurse answers a call from the patient, wherein a first one of the headwall state values defines an on or off state of the nurse answer light.

13. The patient support apparatus of claim 12 further comprising a nurse call light adapted to be illuminated when the patient calls a remotely positioned nurse, wherein a first one of the headwall state values defines an on or off state of the nurse call light.

14. The patient support apparatus of claim 12 further comprising a cable detector adapted to detect a presence of a cable connecting the patient support apparatus to a wall outlet defined in the headwall, wherein a first one of the headwall state values defines an enabled or disabled state of the cable detector.

15. The patient support apparatus of claim 13 further comprising a memory in which is stored at least one of the following: a configuration setting of the headwall unit, wherein a first one of the headwall state values defines a configured state or an unconfigured state of the headwall unit; a television setting of a television coupled to the headwall unit, wherein a first one of the headwall state values defines a type of the television; or a cable setting of a cable extending between the headwall unit and a wall outlet defined in the headwall, wherein a first one of the headwall state values defines a connected state or an unconnected state of the cable.

16. The patient support apparatus of claim 13 further comprising a memory in which is stored an interlock setting of a pair of pins of the headwall unit, wherein a first one of the headwall state values defines an electrically shorted state or unshorted state of the pair of pins.

17. The patient support apparatus of claim 13 further comprising a third wireless transceiver adapted to wirelessly communicate with a fourth wireless transceiver positioned onboard the headwall unit when the patient support apparatus is positioned adjacent to the headwall unit, wherein the communication link is established when both the first wireless transceiver and the third wireless transceiver are in communication with the second and fourth wireless transceivers, respectively, onboard the headwall unit, and wherein the first wireless transceiver is a radio frequency transceiver and the third wireless transceiver is an infrared transceiver.

18. The patient support apparatus of claim 13 wherein the controller is further adapted to send audio signals to the headwall unit via the first wireless transceiver that are to be forwarded by the headwall unit to a wall outlet integrated into the headwall.

19. The patient support apparatus of claim 13 further comprising an exit detection system adapted to detect a patient exit from the support surface, wherein the controller is further adapted to send an exit alert message to the headwall unit via the first wireless transceiver that is to be forwarded by the headwall unit to a wall outlet integrated into the headwall.

20. The patient support apparatus of claim 13 further comprising: a memory in which is stored a first unique identifier that uniquely identifies the patient support apparatus; and a network transceiver adapted to communicate with a computer network of the healthcare facility; wherein the controller is further adapted to receive a second unique identifier of the headwall unit from the headwall unit and to forward both the first and second unique identifiers to a server on the computer network via the network transceiver.
